ANDRÉ LUIZ BARBOSA NUNES DA CUNHA

Tenured Associate Professor of Civil Engineering | AI & Sustainability in Transport
University of São Paulo (USP) • São Carlos School of Engineering (EESC) Department of Transport Engineering (STT) São Carlos, SP, Brazil

alcunha@usp.br | +55 16 98119 2339 | [Google Scholar] | [ORCID] | [LinkedIn] | [LinkTree]

Professional Summary

Tenured researcher and Technical Lead with 12+ years of experience at the intersection of Intelligent Transportation Systems (ITS), Artificial Intelligence, and transport safety. Director of the ITS Research Group at USP-EESC, with a sustained research agenda spanning computer vision for traffic monitoring, traffic simulation, graph-based network vulnerability analysis, and digital twins for highway concessions. Proven track record of translating academic R&D into operational safety standards for major Brazilian highway concessionaires (CCR, ARTERIS). Secured approximately **USD 3.38 million (BRL 18.6M)** in competitive grants and industry contracts. Global academic reach includes visiting appointments in Australia (UoM), Germany (TUM), Croatia (UNIZG), and Portugal (UMinho).

Areas of Expertise

- **Intelligent Transportation Systems (ITS):** traffic monitoring, connected/autonomous vehicles, variable speed limits, smart intersections
- **Artificial Intelligence & Computer Vision:** deep learning (PyTorch, TensorFlow), large multimodal models (LMMs), OpenCV pipelines for vehicle detection and classification
- **Traffic Simulation & Modelling:** PTV VISSIM, AIMSUN, SUMO, MATSim, TSIS-CORSIM — calibration, validation, API scripting
- **Network Analysis & Digital Twins:** graph neural networks (GNNs), urban road network vulnerability, prescriptive digital twins for highway management
- **Highway Safety:** truck escape ramp design, heavy vehicle operations, geometric design, operational speed validation
- **Data Science:** Python, R, C++, Julia, SQL — applied to crash prediction, travel behaviour, spatial equity

EDUCATION

1. **Ph.D. in Transportation Engineering** ..... Nov. 2013
   University of São Paulo (USP), São Carlos School of Engineering (EESC), Brazil
   - Thesis: "*Automatic system for vehicular traffic parameters using OpenCV*"
   - DOI: 10.11606/T.18.2013.tde-19112013-165611
   - Funded by CNPq (National Council for Scientific and Technological Development), Brazil.
2. **M.Sc. in Transportation Engineering** ..... Oct. 2007
   University of São Paulo (USP), São Carlos School of Engineering (EESC), Brazil
   - Thesis: "*Evaluation of performance measurement impact on truck passenger car equivalents*"
   - DOI: 10.11606/D.18.2007.tde-27112007-094400
   - Funded by CNPq, Brazil.
3. **B.S. in Civil Engineering** ..... Feb. 2004
   Federal University of Mato Grosso do Sul (UFMS), Brazil
   - *GPA*: 3.79/4.00 → (9.5/10.0), (Highest Honors)

ACADEMIC APPOINTMENTS

Tenured Associate Professor ..... 2014 – present
University of São Paulo (USP), Dept. of Transport Engineering, São Carlos, Brazil.

- Director of the Intelligent Transportation Systems (ITS) Research Group.
- Tenure awarded 2021; promoted to Senior Faculty status (MS-3.2).
- Elected: Deputy Coordinator of Graduate Program, Department Council, Graduate Coordinating Committee (CCP-ET), Center for Educational Technology in Engineering (CETEPE), Research and Innovation Committee (CPqI), Culture & Extension Committee (CCEx)

Research Scientist & Laboratory Specialist ..... 2013 – 2014
University of São Paulo (USP), Department of Transport Engineering (STT), São Carlos, Brazil.

Assistant Professor 2010
São Paulo State University (UNESP), College of Engineering of Bauru (FEB), Bauru, Brazil.

INTERNATIONAL MOBILITY

- **University of Melbourne** (Australia) - Visiting Researcher, CAPES-Print (2020)
- **University of Zagreb** (Croatia) - Visiting Lecturer, ERASMUS+ (2018, 2022)
- **University of Minho** (Portugal) - Visiting Professor, CAPES-FCT (2017)
- **Technical University of Munich** (Germany) - Visiting Professor, FAPESP-BAYLAT (2016, 2017)

RESEARCH AGENDA - Sustainability & Innovation

- **Cognitive Mobility in Transportation Engineering** (CNPq PQ, 2026–2029). Framework for generative AI applied to data imputation and prescriptive digital twins for transport networks, building on prior work in computer vision and graph theory.
- **RESILIUS — Innovation Network for Resilient, Inclusive and Sustainable Cities** (MCTI/FINEP/FNDCT, 2025–2028). *Co-ordinator of RESILIUS_IA* — Urban Digital Twin platform based on dynamic graphs, integrating real-time data streams (IoT, drones, hydrological sensors, traffic cameras) to support public managers in disaster response, infrastructure monitoring, and mobility equity decisions.
- **Pavement Recycling & Decarbonization (SDG 9 & 12)**: Co-PI on FAPESP USD 1.45M initiative. Data-driven modelling and optimisation for recycled pavement materials; engineering technical protocols for net-zero infrastructure.
- **Urban Climate Resilience & Social Equity (SDG 11 & 13)**: Assessing multimodal network vulnerability to extreme floods using graph theory. Designing fail-safe transport systems for marginalised communities.
- **Next-Gen AI for Global Safety Standards (SDG 3.6)**: Deep learning and large multimodal models for real-time traffic anomaly detection. Scaling national concessionaire work to global operational standards.
- **Digital Twins & International Policy (SDG 17)**: Digital twins for highway concessions integrating Edge AI and sensor fusion for smart city policies.

RESEARCH FUNDING & GRANTS (~USD 3.38 Million Total)

Competitive Research Grants (Peer-Reviewed)

- **2026-2029:** CNPq Research Productivity - Cognitive Mobility in Transportation Engineering. PI ~USD 8,000
- **2025–2030:** FAPESP CCD Grant — "Pavement Recycling as a Pillar of Decarbonisation." Co-PI. ~USD 1.45M
- **2023–2028:** FAPESP CEPID "IARA" — AI Recreating Environments. Research Collaborator. ~USD 1.8M
- **2023–2026:** CNPq Consolidated Groups — Traffic modelling for smart/connected cities. PI. ~USD 13,900
- **2023–2026:** CNPq Research Productivity — AI tools for urban mobility. PI. ~USD 7,200
- **2021:** Uni. Zagreb — Innovative Control Strategies for Sustainable Mobility. Co-PI. ~USD 9,500
- **2019:** CAPES-Print — Visiting Professorship, University of Melbourne. PI. ~AUD 150,000
- **2018–2022:** CNPq Universal — Image-based truck axle detection and classification. PI. ~USD 2,700
- **2018:** Uni. Zagreb — Deep learning in intelligent traffic control. Co-PI. ~USD 10,100

- **2013–2016:** CAPES/FCT — Promoting sustainable and safe urban mobility. Co-PI. ~USD 3,600

Industry Contracts & Translational Research

- **2025:** CCR Group (RioSP) — Redesign truck escape ramp BR-116 (Via Dutra). PI. ~USD 32,700
- **2023:** CCR Group (RioSP) — Evaluate truck escape ramp BR-116 (Via Dutra). PI. ~USD 16,300
- **2019:** ARTERIS (ALS) — Site optimisation for truck escape ramps BR-376. PI. ~USD 27,200

RESEARCH SUPERVISION & MENTORING

Summary 60+ researchers directed. 6 PhD (3 completed, 3 in progress) | 15 MSc (12 completed, 3 in progress) | 18 scientific initiation (16 completed) | 25 capstone projects completed

Completed PhD (4)

1. T.V. Louro (2026): "*Estimating the impacts of electric bicycles on accessibility to jobs and spatial equity.*"
2. L.A. Marcomini (2025): "*Classification of trucks by axles using Deep Learning and a Large Multimodal Model*"
3. A.B. Morelli (2025): "*Analysis of Flood Vulnerability in Brazilian Urban Networks Using Graph Theory Tools.*"
4. G.J.M. Oliveira (2024): "*Toward a Data-driven Framework to Analyse Temporal and Spatial Distributions of Crashes*"

PhD in Progress (2)

1. E.R. Ribeiro: "*Analysis of motorcyclist' behaviour on urban segments: Relationship between riding patterns and rider profile*"
2. P.H.C. Caliari: "*Evaluation of causal inference and spatial effects on travel behavior*"

Completed MSc (12)

1. C.E. Ribeiro (2024): "*Evaluation of deep neural networks for vehicle detection in satellite images*"
2. P.Y. Matsumoto (2022): "*Calibration of Cellular Automata model for simulation of the traffic flow behavior in São Paulo roads*"
3. H.S. Stefani (2021): "*Urban roadway traffic flow prediction from crowdsourced speed data*"
4. A.D.Bosco Jr (2020): "*Usability of Points of Interest and network centralities of collaborative maps for trip attraction analysis: case study of Curitiba*"
5. A.B. Morelli (2019) "*Exploratory analysis of resilience in urban road networks*"
6. B. Kuramoto (2019): "*Data exploration of collaborative maps in evaluations of Brazilian urban morphologies*"
7. A.B. Felicio (2019): "*Evaluation of the behavior of motorcyclists through the video image processing system*"
8. N.R. Panice (2018): "*Truck axle detection automatic method based on images*"
9. M.M. Thebit (2018): "*Reconstruction of a synthetic O/D matrix using traffic data available on the web*"
10. L.A. Marcomini (2018): "*Automatic identification of traffic behavior using video images*"
11. G.J.M. Oliveira (2018): "*Calibration of speed-flow relationship for freeways and multilane highways*"
12. E.R. Ribeiro (2017): "*Exploratory method analysis using Wavelet to detect patterns and anomalies in traffic history data*"

MSc in Progress (4)

1. A.V. Costa "*Accessibility to Social Services in Belo Horizonte-MG*"
2. M.E.S. Michelini "*Development of an Emission Estimation Model for Brazilian Cases*"
3. R.O.F.P. Oliveira "*Urban resilience assessment through a comparative study of flood prediction methods*"
4. M.C. da Silveira "*Urban vulnerability indicator based on historical HAND data.*"

Alumni Placement

Graduates placed at national regulatory agencies (ANTT, ARTESP) in Brazil, leading transport consultancies in Brazil and abroad, and the financial sector.

TEACHING PORTFOLIO (USP, 2014-present)

Data Science & Computational Engineering

- **Multivariate Data Analysis & Applied Statistics (Graduate):** AI techniques (neural networks, decision trees, genetic algorithms), R programming. Students publish from course datasets.
- **Computational Tools for Civil Engineering (Undergraduate):** New elective I designed. 20+ students/year. Spreadsheet programming, algorithms, AutoCAD, Civil 3D, GIS.

Core Transport Engineering & Simulation

- **Fundamentals of Transportation Engineering (Undergraduate):** Inverted classroom, project-based learning, 50+ students. Student evaluations: 4.5/5.0.
- **Traffic Engineering & Simulation (UG/Graduate):** VISSIM, AIMSUN, SUMO, TSIS-CORSIM. Industry-ready skills.

Logistics & Green Infrastructure

- **Logistics & Supply Chain Management:** Completely redesigned with AI logistics, GIS route planning, Green Logistics.
- **Airports & Waterways:** Large compulsory modules, 50+ students, theory and operational exercises.

Capstone & Mentorship

- Supervised 25+ final-year engineering capstone projects and 15+ internship placements.

PUBLICATIONS

20 peer-reviewed journal articles | 36 refereed conference proceedings | 3 manuscripts under review.

Students mentored by Prof. A.L. Cunha are denoted by underlining.

Submitted Manuscripts

1. SALINAS, K., BARELLA, V., **CUNHA, A.L.**, OLIVEIRA, G.M., VIERA, T., NONATO, L.G. (2025) "ORDENA: ORigin-DEstiNAtion data exploration". *IEEE Transactions on Visualization and Computer Graphics* <arXiv:2510.18278>

Peer-Reviewed Journal

1. LOURO, T. V., TOMASIELLO, D. B., GRIGOLON, A. B., **CUNHA, A. L.**, GEURS, K. T. (2026). "Integrating e-bikes and bikesharing with public transit: Evaluation of impacts on job accessibility and spatial equity". *Journal of Urban Mobility*.
2. LOURO, T. V., GRIGOLON, A. B., **CUNHA, A. L.**, GEURS, K. T. (2025). "Cycling accessibility and equity: A systematic literature review". *Transport Reviews*. 10.1080/01441647.2026.2664409
3. LOURO, T.V.; GRIGILON, A.B.; TIRACHINI, A.; **CUNHA, A.L.**; GEURS, K.T. (2026) "How Do E-Bikes Measure Up? Analyzing Speed Differences and Network Impacts of São Paulo's Bikesharing System". *Transportation*. DOI: 10.1007/s11116-026-10724-y
4. LOURO, T.V.; GRIGILON, A.B.; **CUNHA, A.L.**; GEURS, K.T. (2025) "E-bikes' impact on job accessibility and equity in São Paulo and Rio". *Transportation Research Part D*. DOI: 10.1016/j.trd.2025.105072
5. DE OLIVEIRA, G.J.M.; LAVIERI, P.S.; **CUNHA, A.L.** (2023) Integrating a non-gridded space representation into a graph neural networks model for citywide short-term crash risk prediction. *Urban Informatics*. DOI: 10.1007/s44212-023-00032-6
6. FLEURY, M.P.; KAMAKURA, G.K.; PITOMBO, C.S.; **CUNHA, A.L.B.N.**; FERREIRA, F.B.; LINS DA SILVA, J. (2023) Assessing and Predicting Geogrid Reduction Factors after Damage Induced by Dropping Recycled Aggregates. *Sustainability*. DOI: 10.3390/su15139942
7. FLEURY, M.P.; KAMAKURA, G.K.; PITOMBO, C.S.; **CUNHA, A.L.B.N.**; LINS DA SILVA, J. (2023) Prediction of non-woven geotextiles' reduction factors for damage caused by the drop of backfill materials. *Geotextiles and Geomembranes*. DOI: 10.1016/j.geotexmem.2023.05.004
8. SILVA, F.A.E.; BESSA JUNIOR, J.E.; COSTA, A.L.; **CUNHA, A.L.**; VELHO, D.M.C.; ANDALICIO, A. (2023) Exploratory analysis of the VISSIM simulation model for two-lane highways. *Engenharia Civil UM*. DOI: 10.21814/ecum.4493
9. SILVA, F.A.; BESSA JUNIOR, J.E.; COSTA, A.L.; **CUNHA, A.L.**; VELHO, D.M.C. (2022) Analysis of no-passing zones to assess the level of service on two-lane rural highways in Brazil. *Case Studies on Transport Policy*. DOI: 10.1016/j.cstp.2021.12.006
10. MORELLI, A. B.; **CUNHA, A.L.** (2021) Assessing vulnerabilities in transport networks: a graph-theoretic approach. *Transportes*. DOI: 10.14295/transportes.v29i1.2250
11. SILVA, F.A.; BESSA JÚNIOR, J.E.; COSTA, A.L.; **CUNHA, A.L.**; ANDALÍCIO, A.F.; DA COSTA VELHO, D.M.; NAZARETH, V.S. (2021) Evaluation of the effect of climbing lanes on segments of two-lane highways. *Transportes*. DOI: 10.14295/transportes.v29i2.2359
12. MORELLI, A.B.; **CUNHA, A.L.** (2021) Measuring urban road network vulnerability to extreme events: An application for urban floods. *Transportation Research Part D*. DOI: 10.1016/j.trd.2021.102770
13. MARTINS, D.O.; OLIVEIRA, G.J.M.; MORAES, F.R.; SILVA, I.; **CUNHA, A.L.** (2020) Geomatics data management system. *Revista Brasileira de Geomática*. DOI: 10.3895/rbgeo.v8n1.10141
14. PIANUCCI, M.N.; PITOMBO, C.S.; **CUNHA, A.L.**; LIMA SEGANTINE, P.C. (2019) Forecasting household travel demand through a sequential method based on synthetic population and artificial neural networks. *Transportes*. DOI: 10.14295/transportes.v27i4.1409
15. OLIVEIRA, J.V.M.; LAROCCA, A.P.C.; ARAUJO NETO, J.O.; **CUNHA, A.L.**; SANTOS, M.C.; SCHAAL, R.E. (2019) Rigid Bridges Health Dynamic Monitoring Using 100 Hz GPS Single-Frequency and Accelerometers. *Positioning*. DOI: 10.4236/pos.2019.102002
16. DE OLIVEIRA, J.V.M.; LAROCCA, A.P.C.; DE ARAÚJO NETO, J.O.; CUNHA, A.L.; DOS SANTOS, M.C.; SCHAAL, R.E. (2019) Vibration monitoring of a small concrete bridge using wavelet transforms on GPS data. *Journal Of Civil Structural Health Monitoring*. DOI: 10.1007/s13349-019-00341-y
17. LINDNER, A.; PITOMBO, C.S.; **CUNHA, A.L.** (2017) Estimating motorized travel mode choice using classifiers: An application for high-dimensional multicollinear data. *Travel Behaviour and Society*. DOI: 10.1016/j.tbs.2016.08.003
18. SOUZA, N.C.; PITOMBO, C.; **CUNHA, A.L.**; LAROCCA, A.P.C.; DE ALMEIDA FILHO, G.S. (2017) Model for classification of linear erosion processes along railways through decision tree algorithm and geotechnologies. *Boletim de Ciências Geodésicas*. DOI: 10.1590/S1982-21702017000100005
19. ANDRADE, G.R.; PITOMBO, C.; **CUNHA, A.L.N.**; SETTI, J.R. (2016) A Model for Estimating Free-Flow Speed on Brazilian Expressways. *Transportation Research Procedia*. DOI: 10.1016/j.trpro.2016.06.032

20. LAROCCA, A.P.C.; ARAUJO NETO, J.O.; BARBOSA, A.C.B.; TRABANCO, J.L.A.; **CUNHA, A.L.B.N.** (2014) Dynamic Monitoring vertical Deflection of Small Concrete Bridge Using Conventional Sensors And 100 Hz GPS Receivers - Preliminary Results. *IOSRJEN Journal of Engineering*. DOI: 10.9790/3021-04920920
21. **CUNHA, A.L.**; SETTI, J.R. (2011) Truck equivalence factors for divided, multilane highways in Brazil. *Procedia: Social and Behavioral Sciences*. DOI: 10.1016/j.sbspro.2011.04.447

Selected Conference Proceedings (36 total: 10 most recent shown)

1. MORELLI, A.B.; ALIZON, G.B.; **CUNHA, A.L.** (2025) Alternative-Route Efficiency in Brazilian Cities: How Flood-Induced Collapse Patterns Differ from Random Blockages. In: XXXIX ANPET – Research and Teaching in Transport Congress, 2025, Goiânia. Proceedings of the 39th ANPET.
2. LOURO, T.V.; ASSIS, L.B.M.; JUNIOR, J.U.P.; **CUNHA, A.L.**; GEURS, K.T (2025) Job Accessibility in the 15-Minute City: A Comparative Analysis of Walking, Cycling, and E-Bikes in Four Brazilian Cities. In: XXXIX ANPET – Research and Teaching in Transport Congress, 2025, Goiânia. Proceedings of the 39th ANPET.
3. ISHIHARA, B.A.; QUINTINO, P.G.; **CUNHA, A.L.**; SETTI, J.R. (2025) Operation of Heavy Vehicles on Long, Steep Downgrades: Brake Thermal Simulation Based on ABNT NBR 10966-2. In: XXXIX ANPET – Research and Teaching in Transport Congress, 2025, Goiânia. Proceedings of the 39th ANPET.
4. RODRIGUES, L.R.; PITOMBO, C.S.; **CUNHA, A.L.**; LAROCCA, A.P.C.; FERRAZ, A.C.P. (2025) Identification of Crime and Traffic Crash Hotspots in the Vicinity of Bus Stops. In: XXXIX ANPET – Research and Teaching in Transport Congress, 2025, Goiânia. Proceedings of the 39th ANPET.
5. DAVOLI, J.P.; PITOMBO, C.S.; **CUNHA, A.L.** (2025) Application of a Random-Forest–Based Variable Selection Method for the Analysis of Post-COVID-19 Active Transportation. In: XXXIX ANPET – Research and Teaching in Transport Congress, 2025, Goiânia. Proceedings of the 39th ANPET.
6. MORELLI, A.B.; **CUNHA, A.L.** (2024) Vulnerability to flooding: how long-trip prevalence reduces the efficiency of alternative routes. In: XXXVIII ANPET – Research and Teaching in Transport Congress, 2024, Florianópolis. Proceedings of the 38th ANPET.
7. MARCOMINI, L.A.; **CUNHA, A.L.** (2023) Truck axle detection using Neural Networks: analysis of the number of images in the training dataset. In: ANPET – Research and Teaching in Transport Congress, 2023, Santos. Proceedings of the 37th ANPET.
8. MORELLI, A.B.; LOURO, T.V.; **CUNHA, A.L.** (2022) Proposal of bikeability indicators from an accessibility perspective: identifying roads best suited for cycle lanes using widely available data. In: XXXVI ANPET – Research and Teaching in Transport Congress, 2022, Fortaleza. Proceedings of the XXXVI ANPET.
9. MORELLI, A.B.; **CUNHA, A.L.** (2021) Pedestrian accessibility: impacts of morphological and demographic characteristics on access to facilities. In: XXXV ANPET – Research and Teaching in Transport Congress, 2021. Proceedings of the XXXV ANPET.

1. BOSCO JUNIOR, A.D.; **CUNHA, A.L.** (2020) Street and zonal scale relationship between network centrality and economic activities: case study in Curitiba, Brazil. In: PLURIS – Luso-Brazilian Congress on Urban, Regional, Integrated and Sustainable Planning, 2020. Proceedings of the 9th PLURIS.

AWARDS & HONORS

- **ANPET Scientific Production Award** (2023) — National Agency for Transport Research & Education
- **Best Professor, Transport Engineering** (2016, 2017) — USP Civil Engineering Academic Secretariat
- **ABCR Innovation Salon Award** (2015) — 9th Brazilian Congress on Highways & Concessions

PROFESSIONAL SERVICES

University of São Paulo

- Deputy Coordinator, Graduate Program in Transportation Engineering
- Elected member, Department Council — faculty hiring, budget, strategic partnerships
- Member, Graduate Program Coordinating Committee (CCP-ET) — admissions, curriculum, scholarships
- Member, Research and Innovation Committee (CPqI) — interdisciplinary grants, industry collaboration
- Board member, Center for Educational Technology (CETEPE) — e-learning strategy
- Member, Culture & Extension Committee (CCEx) — pedagogical modernisation

Editorial & Peer Review

- Reviewer: *Transportation Research Part E*, *Engineering Applications of AI (EAAI)*, *Sustainable Cities and Society*
- Technical Committees: Transportation Research Board (TRB), IEEE ITSS, ANPET

Industry Engagement

- CCR Group & ARTERIS — national safety standards for truck escape ramps on Brazilian federal highways
- TECTRAN — Minas Gerais State Logistics Plan (PELT)
- Media: research featured on Rede Globo's *Jornal Hoje* (national broadcast)